

Yash Rawal

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Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms, hierarchical RAG architectures, and automated intelligence pipelines across 120+ organizations. Specialized in Python, LLM orchestration (LangChain, LangGraph, ReAct loops), vector search, and model evaluation control planes. Active contributor to Spring AI, bridging generative AI and model fine-tuning with resilient production systems.

Skills

Languages & Backend: Python, SQL, Java, FastAPI, Spring Boot, PostgreSQL, REST APIs

AI & ML Engineering: Agentic AI, RAG & Vector Search, Multi-Agent Systems, Model Training & SFT, Context Graphs

AI & ML Frameworks: LangChain, LangGraph, Spring AI, PyTorch, vLLM, LlamaIndex, CrewAI, GROBID

Platforms & Models: Azure OpenAI, Claude, Amazon Bedrock, Google Vertex AI, Qwen, Meta Llama

Infrastructure & Security: Docker, Kubernetes, Linux, CI/CD, LLM-as-Judge, Red Teaming, PII Sanitization

Work Experience

Bigscal Technologies Pvt. Ltd. (Client: TalenTeam Ltd., UK)	
AI/ML Engineer	Oct 2024 - Present
Remote	
- Deployed a production-scale agentic platform and multi-agent systems serving 1M+ active ERP users across 120+ enterprise tenants. - Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by 3x and token consumption by 70%. - Trained domain-specific Qwen2.5-7B models on A100 GPUs and fine-tuned E5-large-v2 embeddings using PyTorch to optimize query intent mapping. - Developed the Prompt Lab evaluation control plane to collect reasoning trajectories and human preference annotations, establishing a scalable Human Feedback (RLHF) Pipeline. - Implemented a hierarchical RAG pipeline indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a 90%+ Recall@10 retrieval rate. - Secured agentic workflows against OWASP LLM vulnerabilities using phantom-token context propagation, runtime tool monitoring, and trained token-classification models for automated PII redaction.	
Bigscal Technologies Pvt. Ltd. (Product: Healthray)	
AI/ML Engineer Intern	Apr 2024 - Sep 2024
Surat, India	
- Engineered clinical summarization pipelines for 116K+ medical records with 4-bit quantized Llama 3, and designed real-time WebSocket audio streaming for Indic ASR (Whisper, Vosk, NeMo). - Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.	

Projects

Modular Agent Harness & Memory System ("Brain") Open Source	
Principal Creator & Architect of an open-source plug-and-play agentic framework designed as modular building blocks for flexible enterprise workflows, implementing a formal ReAct loop with step budgets, vector tool routing, and dynamic MCP settings caching.	
Java, Spring Boot, PostgreSQL, WebSockets	
Spring AI (spring-projects/spring-ai) Open Source	
Merged PR #5711 to propagate reasoningContent from AWS Bedrock and OpenAI-compatible models, and resolved Converse API prompt-caching TTL latency consistency.	
Java, Spring Boot, Python	
Mini Language Model (~85M Parameter Architecture) Personal Research	
Pretrained an ~85M-parameter transformer to ~1.43 validation loss on TinyStories; engineered FragmentStream chunked attention to eliminate $O(T^2)$ VRAM overhead on commodity GPUs and custom Triton kernels with KV caching for low-latency inference.	
Python, PyTorch, CUDA, Triton, Tesla P100/T4	

Education

Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University	2022 - 2024
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Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900)	Microsoft
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